

Indirect Effect Estimation with Error-Correction Models in Multilevel Mediation Analysis

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Introduction

Multilevel mediation analysis is widely used in social science research. Among its variations, the 1-1-1 multilevel mediation model—where predictor (X), mediator (M), and outcome (Y) are all measured at the within-level—is commonly applied to investigate both within- and between-level effects, especially in educational research where contextual effect studies are prevalent. However, psychological constructs are typically measured with some degree of measurement and sampling error, which can bias parameter estimates and reduce estimation efficiency. To control these errors, doubly latent multilevel structural equation modeling (DL-MSEM) can be applied by specifying latent constructs through confirmatory factor analysis at both the within- and between-levels. Yet, the complexity of DL-MSEM often leads to estimation difficulties, including non-convergence and negative variances, especially with small samples or low intraclass correlation (ICC) (Li & Beretvas, 2013; Lüdtke et al., 2011). Bayesian estimation has been proposed as a potential remedy, given that appropriate priors can stabilize estimation in small samples (Depaoli & Clifton, 2015; McNeish, 2016). However, the accuracy of prior information cannot always be ensured in practice, and the use of software defaults or inaccurate priors can lead to increased bias and reduced estimation accuracy. Consequently, researchers who lack confidence in their priors may hesitate to adopt Bayesian models and instead rely on traditional manifest approaches without latent aggregation. Yet, the estimation performance of these approaches has not yet been examined simultaneously.

Research Purpose

This study aimed to evaluate the performance of several 1-1-1 multilevel mediation modeling approaches for estimating the between-level indirect effect. Using a Monte Carlo simulation, this study:

- 1) examined the conditions under which frequentist DL-MSEM exhibits estimation issues, and
- 2) compared frequentist models with varying degrees of error correction to Bayesian DL-MSEM under different prior specifications, thereby identifying viable alternatives when the frequentist DL-MSEM

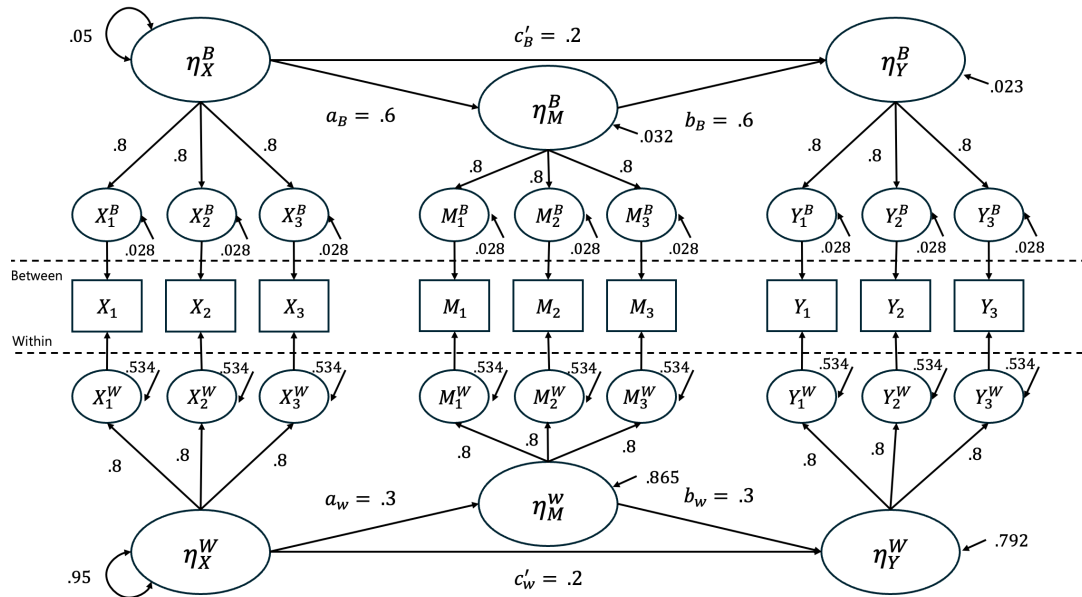
cannot be applied. Specifically, the study aims to determine the extent to which simpler frequentist alternatives, such as the partial error-correction model, are more biased or less accurate than Bayesian approaches with default or inaccurate priors. The findings are expected to provide methodological insights and guidance for applied researchers in selecting appropriate approaches for 1-1-1 multilevel mediation analysis.

Methods

Data were generated from a 1-1-1 DL-MSEM (**Figure 1**) with random intercepts and fixed slopes. The manipulated conditions included the intraclass correlation coefficient (ρ_{ICC} : .05, .20), number of clusters (J : 20, 50, 100), cluster size (n : 10, 20), standardized factor loadings (λ_{std} : .6, .8), and between-level indirect effect sizes ($a_B b_B$: .01, .09, .36), resulting in 72 total conditions. For each condition, 500 datasets were generated and analyzed with *Mplus* 8.3 (Muthén & Muthén, 1998-2017) and the *MplusAutomation* package in R 4.5.1. (Hallquist & Wiley, 2018; R Core Team, 2025).

Figure 1. 1-1-1 DL-MSEM model with population values

($ICC = .05$, $\lambda_{std} = .8$, $a_B b_B = .36$)



Three frequentist and five Bayesian multilevel mediation models were estimated and compared. The three frequentist models differed in the level of error correction: the doubly manifest multilevel mediation model (**Figure 2**), in which all constructs are treated as manifest variables by averaging across items and individuals within a group; the sampling error correction multilevel mediation model (**Figure 3**), in which sampling error is corrected by decomposing the within-level manifest composites into latent components; and DL-MSEM, in which both sampling and measurement errors are corrected via latent aggregation derived from multilevel confirmatory factor analysis (Lüdtke et al., 2011). When

estimating DL-MSEM, cross-level factor loading invariance constraints were imposed (Jak, 2019). For the frequentist models, 95% confidence intervals of the between-level indirect effect were computed using the Monte Carlo method with 20,000 random draws in R, to account for the asymmetric distribution of the indirect effect (Preacher & Selig, 2012; Selig & Preacher, 2008).

Figure 2. 1-1-1 Doubly manifest multilevel mediation model

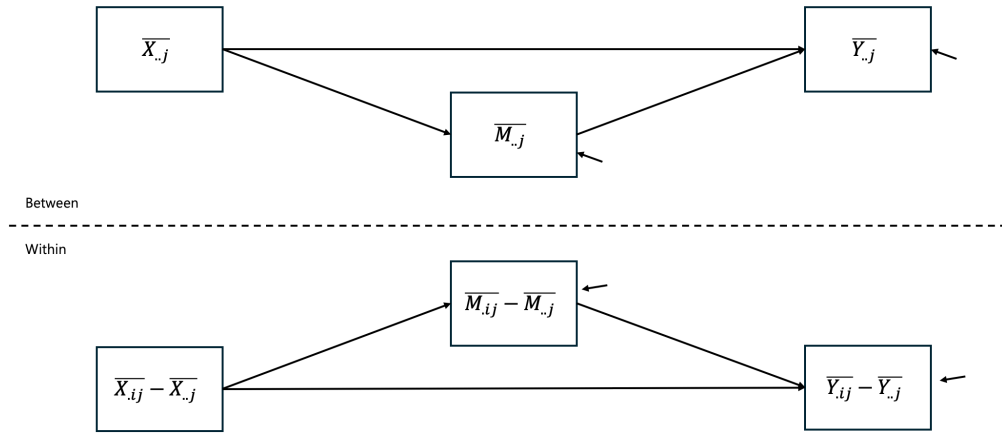
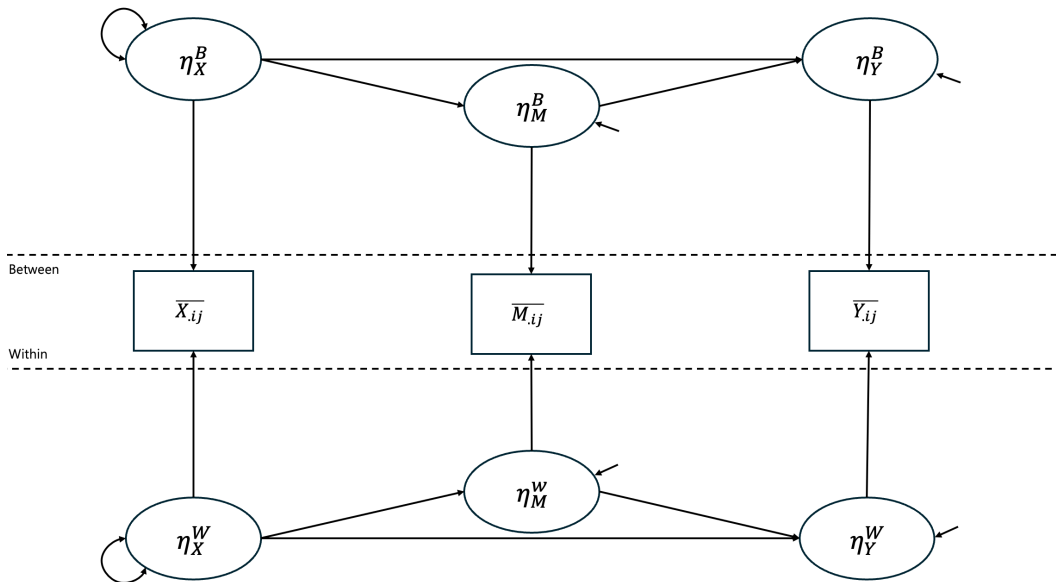


Figure 3. 1-1-1 Sampling error correction multilevel mediation model



For Bayesian estimation, five DL-MSEM models were analyzed, each employing a distinct prior specification. In the first model, non-informative priors were applied using the default settings in *Mplus*: intercepts, factor loadings, and regression coefficients followed $N(0, 10^{10})$, and variance and covariance parameters followed $IG(-1, 0)$, both representing diffuse priors. The remaining four models applied

informative priors to the regression coefficients: *weakly informative inaccurate*, *informative inaccurate*, *weakly informative accurate*, and *informative accurate*. All regression coefficients were assumed to follow a normal distribution, with accurate priors specified by setting the mean hyperparameters equal to the true population values, and inaccurate priors defined by shifting the mean hyperparameters two standard deviations (2 SD) below the true population values to examine their impact on the posterior distributions. The variance hyperparameters determined the degree of prior precision, with values of 0.01 for informative priors and 0.1 for weakly informative priors. Priors for between-level variance parameters were set as $IG(0.1, 0.1)$ to pool the variance away from zero, while within-level variance parameters were set as $IG(0.001, 0.001)$ (Zitzmann et al., 2015, 2016). Bayesian DL-MSEM models were estimated within an MCMC framework using the Gibbs sampler to generate posterior distributions, running two chains for up to 50,000 iterations, with the first half treated as the burn-in phase (Asparouhov & Muthén, 2010; Geman & Geman, 1984). The posterior median was used as the point estimate, and 95% credible intervals were computed.

Model performance was evaluated using convergence rate, relative percentage bias, root mean square error (RMSE), coverage, and non-null detection rate for the between-level indirect effect.

Results

Simulation results showed that when the number of clusters was small ($J = 20$), the frequentist DL-MSEM failed to converge or yielded inadmissible solutions across all replications. Even with moderate or large numbers of clusters ($J = 50, 100$), the frequentist DL-MSEM continued to report estimation issues, though performance improved with larger ICCs, higher standardized factor loadings, and increased sample sizes.

In conditions where the frequentist DL-MSEM suffered from substantial non-convergence and inadmissible solutions—especially with a small number of clusters—only Bayesian DL-MSEM with priors specified by accurate mean and relatively small variance hyperparameters achieved acceptable performance. However, these informative priors imply strong certainty regarding model parameters, which will not always be ensured in applied research. When weakly informative priors were imposed by increasing variance hyperparameters from 0.01 to 0.1, the model showed notable negative bias, which improved toward acceptable levels as ICC increased to .20 and the number of clusters to $J = 50$. At the same time, larger variance hyperparameters induced a wider range of parameter space, leading to reduced non-null detection rates at medium effect sizes, though these rates also improved at larger effect sizes with higher ICC and more clusters.

Regarding *Mplus* default priors—which correspond to noninformative priors that approximate infinite variance hyperparameters—the result showed performance patterns similar to those of weakly

informative accurate priors, occasionally yielding slightly lower bias. However, the default priors exhibited substantial estimation inaccuracy, reflected in large RMSE values that were sometimes even higher than those observed for models with inaccurate priors. This likely indicates a bias-variance trade-off resulting from the broader parameter distributions associated with diffuse priors.

For the frequentist alternatives, only DL-MSEM produced acceptable bias across the simulated conditions when the model converged, especially under higher ICC values. However, it also showed the largest RMSE values among the three frequentist models. In terms of RMSE, the doubly manifest model exhibited the highest accuracy, and the sampling error-correction model occasionally produced the lowest RMSE under high ICC and large numbers of clusters. For instance, under the condition of $\rho_{ICC} = .20$, $\lambda_{std} = .8$, $a_B b_B = .36$, $n = 20$, and $J = 50$, the sampling error-correction model demonstrated slightly lower RMSE and bias than the doubly manifest model. However, unlike the error-correction models, the doubly manifest model did not achieve acceptable coverage across conditions under medium and large effect sizes, indicating its limited applicability in practice.

When comparing the frequentist alternatives with Bayesian models, the doubly manifest and sampling error-correction models showed performance comparable to that of Bayesian DL-MSEM with inaccurate priors, performing particularly better under high standardized factor loadings and larger numbers of clusters. This was most apparent in terms of bias, and for the sampling error-correction model, in coverage rates as well, which were higher than those of the frequentist doubly manifest model and Bayesian models with inaccurate priors. However, these frequentist alternatives still yielded lower performance compared with Bayesian models using accurate priors.

Therefore, when frequentist DL-MSEM encounters estimation issues, researchers who lack confidence in specifying prior information for model parameters may consider simpler frequentist alternatives, such as the sampling error-correction model, particularly under high item reliability (i.e., high standardized factor loadings), high ICC, and when at least 50 clusters are available. Although these simpler models tend to produce negatively biased estimates, they can still provide more accurate results than DL-MSEM, even when the latter successfully converges.

Implications

This simulation study aimed to address the challenges in estimating between-level indirect effects in 1-1-1 multilevel mediation analysis within the structural equation modeling (SEM) framework. Specifically, it reconfirmed the conditions under which DL-MSEM encounters estimation issues and examined the performance of Bayesian DL-MSEM under various prior specifications, along with the applicability of the sampling error-correction model under specific conditions. Since previous research providing practical guidance for applied researchers on 1-1-1 multilevel mediation with multiple

indicators has been limited, the present study offers important insights by demonstrating that estimation performance improves with higher item reliability (i.e., standardized factor loadings), as well as with larger ICC values and sample sizes highlighted in previous literature. By focusing on the bias-variance trade-off, this study demonstrated the relative advantages and drawbacks of each modeling approach, thereby guiding researchers in selecting appropriate methods for their substantive research questions.

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